

The Master Command Log

1. System Prep & Repository Setup

- a. `sudo apt update && sudo apt upgrade -y`
- b. `sudo apt install wget curl gnupg apt-transport-https jq -y`
- c. `wget -qO - https://artifacts.elastic.co/GPG-KEY-elasticsearch | sudo gpg
--dearmor -o /usr/share/keyrings/elasticsearch-keyring.gpg`
- d. `echo "deb [signed-by=/usr/share/keyrings/elasticsearch-keyring.gpg]
https://artifacts.elastic.co/packages/8.x/apt stable main" | sudo tee
/etc/apt/sources.list.d/elasticsearch-8.x.list`
- e. `sudo apt update`

2. Elasticsearch Setup

- a. `sudo apt install elasticsearch -y`
`sudo nano /etc/elasticsearch/elasticsearch.yml`
 - i. `# Inside nano: Changed xpack.security.enabled: true to false``sudo service elasticsearch start`

3. Kibana Setup

- a. `sudo apt install kibana -y`
`sudo service kibana start`

4. Suricata Setup & Interface Tuning

- a. `sudo add-apt-repository ppa:oisf/suricata-stable -y`
`sudo apt update`
`sudo apt install suricata -y`
`sudo suricata-update`
`sudo nano /etc/suricata/suricata.yaml`
 - i. `# Inside nano: Changed the af-packet interface from eth0 to lo (loopback)
to catch internal lab traffic.`

5. Custom Security Rules

- a. `sudo mkdir -p /etc/suricata/rules`
- b. `sudo nano /etc/suricata/rules/local.rules`
 - i. We pasted the following rules into the file:
 - `alert tcp any any -> any 22 (msg:"LOCAL SSH Brute Force Attempt"; flow:established,to_server; content:"SSH-"; threshold:type both, track by_src, count 5, seconds 60; classtype:attempted-login; sid:1000001; rev:1;)`
 - `alert dns any any -> any 53 (msg:"LOCAL Suspicious DNS Query (malicious.com)"; dns.query; content:"malicious.com"; nocase; classtype:bad-unknown; sid:1000002; rev:1;)`
 - `alert tcp any any -> any any (msg:"LOCAL Nmap SYN Scan Detected"; flags:S; window:1024; threshold:type both, track by_src, count 10, seconds 10; classtype:attempted-recon; sid:1000003; rev:1;)`
- c. `sudo service suricata restart`

6. Filebeat Integration & The Module Fix

- a. `sudo apt install filebeat -y`
`sudo filebeat modules enable suricata`
`sudo filebeat setup -e`
 - i. # The fix to ensure Filebeat read the eve.json file:
- b. `cat << 'EOF' | sudo tee /etc/filebeat/modules.d/suricata.yml`
`- module: suricata`
`eve:`
`enabled: true`
`var.paths: ["/var/log/suricata/eve.json"]`
`EOF`
- c. `sudo service filebeat start`

7. Attack Simulation Tools

- a. `sudo apt install nmap -y`
`sudo nmap -sS -p- -T4 localhost`
- b. `for i in {1..6}; do ssh -o ConnectTimeout=1 fakeuser@localhost; done`

Detection & Visibility Implementation

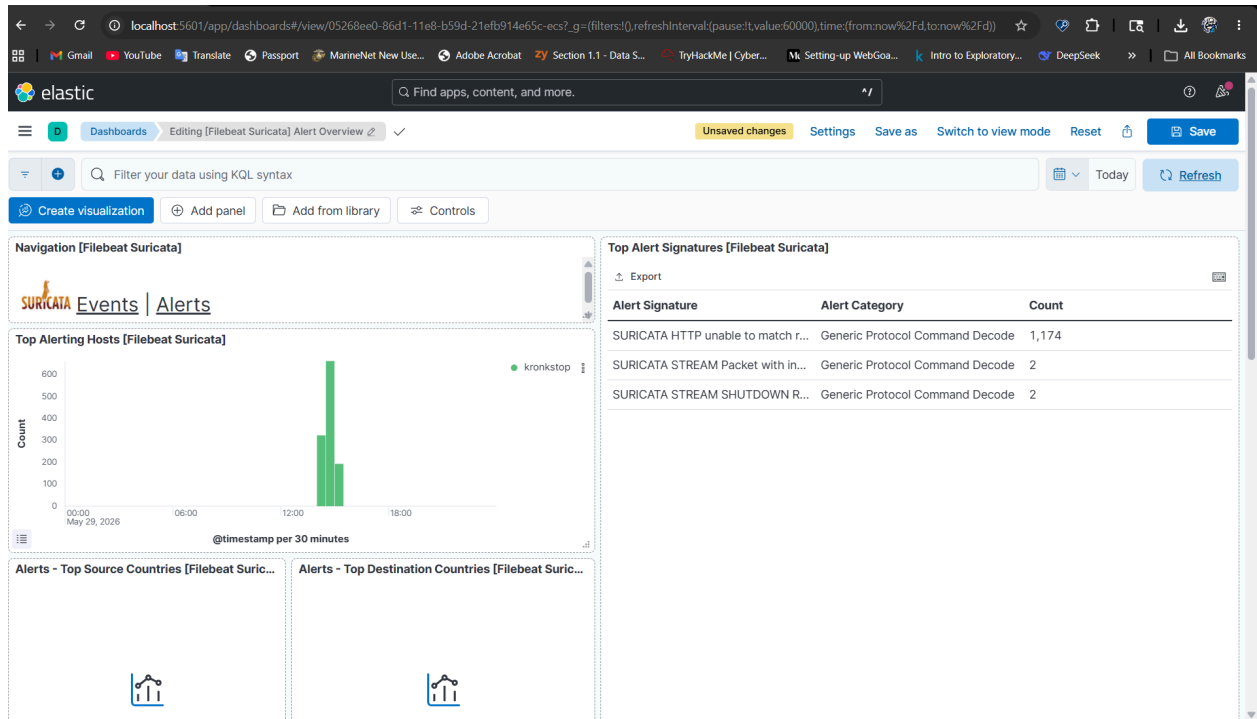
1. Architecture and Setup For the monitoring and detection part of the network lab, we set up a standalone Ubuntu VM to run our open-source security stack. I ended up choosing Suricata as the Intrusion Detection System (IDS) instead of Snort. Suricata handles multi-threading better out of the box, and its default JSON (**eve.json**) log output makes it way easier to feed into modern log aggregators.

To actually get visibility into the network traffic, I installed the ELK Stack (Elasticsearch and Kibana) locally on the VM. I used Filebeat as the log shipper to connect them, turning on the Suricata module so it knew how to parse the raw JSON alerts and push them straight into Elasticsearch. From there, I used Kibana to visualize everything through the pre-built Filebeat Suricata Alert Overview dashboard. One specific tweak during the configuration was tuning Suricata to monitor the loopback interface (**lo**)—this was necessary to make sure it could actually catch the internal test attacks we were running within the virtual environment.

2. Rule Tuning and Rationale To make sure our lab could detect common attacks, we wrote three custom detection rules and loaded them into Suricata's **local.rule** file:

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- **SSH Brute Force Detection:** We designed this to catch credential-stuffing attacks targeting the SSH server. The rule tracks the source IP and fires off an alert if it sees 5 connection attempts hit port 22 within a 60-second window.
- **Suspicious DNS Queries:** This acts as an Indicator of Compromise (IoC). It detects if an internal machine tries to resolve a known malicious domain, which is a classic sign of malware trying to phone home to a Command and Control (C2) server.
- **Nmap SYN Scan Detection:** This rule is built to catch an attacker's early recon efforts. It watches for a quick burst of TCP SYN packets (10 within 10 seconds) from a single source that don't finish the three-way handshake, which is the hallmark of a stealth scan.
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Alerts [Filebeat Suricata]

1,178 documents

☐	@timestamp	host.name	suricata.eve.flow_id	source.ip	source.port	destination.ip	destination.port	source.geo.country_iso_code	destination.geo.country_iso_code
☐	✓ May 29, 2026 @ 15:07:18.519	kronkstop	208121360613559	127.0.0.1	9200	127.0.0.1	39128	-	-
☐	✓ May 29, 2026 @ 15:07:17.509	kronkstop	208121360613559	127.0.0.1	9200	127.0.0.1	39128	-	-
☐	✓ May 29, 2026 @ 15:07:17.441	kronkstop	208121360613559	127.0.0.1	9200	127.0.0.1	39128	-	-
☐	✓ May 29, 2026 @ 15:07:17.422	kronkstop	208121360613559	127.0.0.1	9200	127.0.0.1	39128	-	-
☐	✓ May 29, 2026 @ 15:07:16.695	kronkstop	208121360613559	127.0.0.1	9200	127.0.0.1	39128	-	-

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